# I am losing weight and sweating all the time: Causes of severe weight loss Thyrotoxicosis and Hypothyroidism


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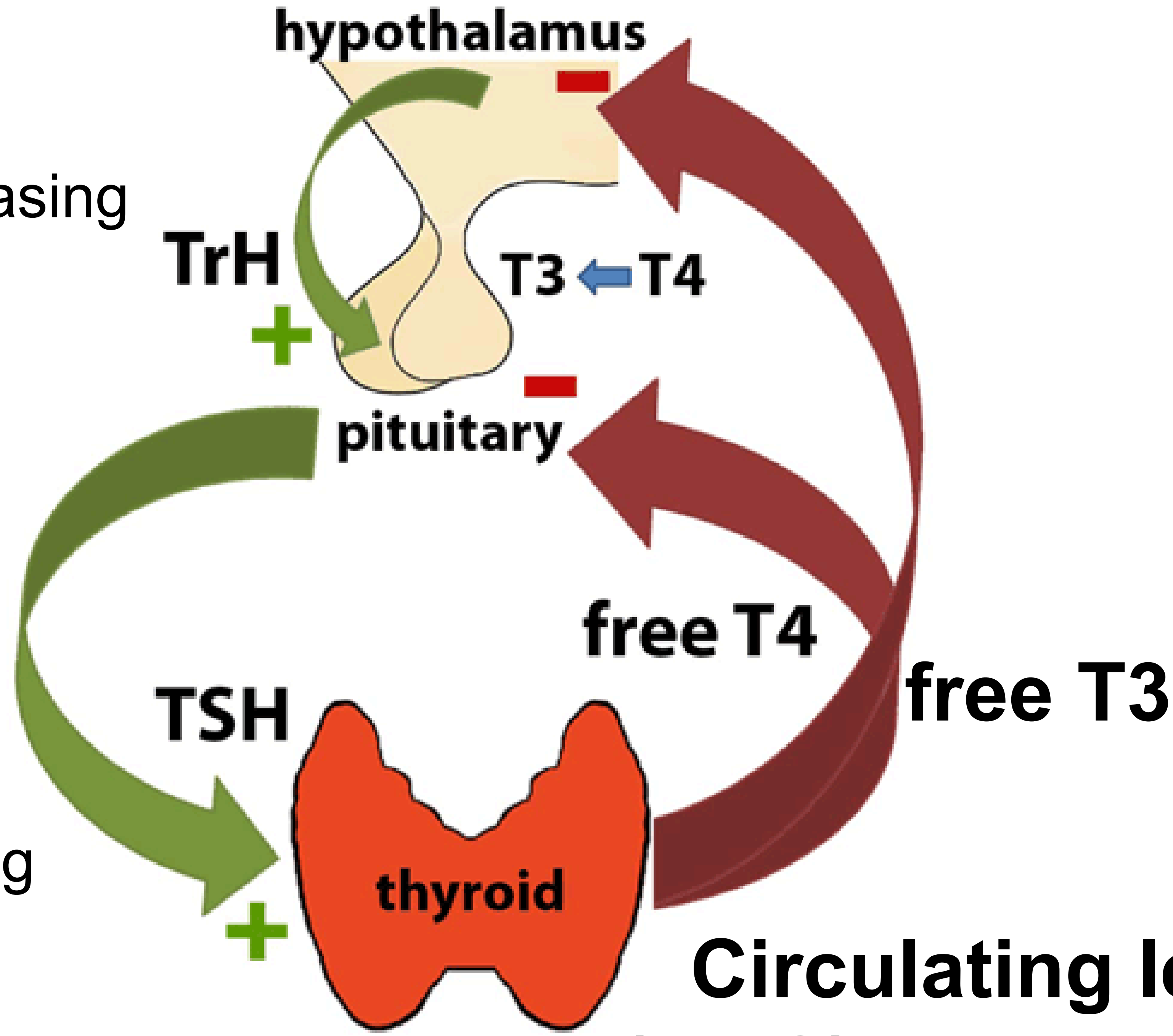
# Case Vignette

- Ms Chan
- F/30, Non smoker
- Good past health
- Family hx of thyroid disease
- Palpitations, easy sweating and weight loss of 10 pounds in 1 month
- Central neck swelling and proptosis
- Blood tests revealed high thyroid hormone and low TSH levels
- ECG showed sinus tachycardia

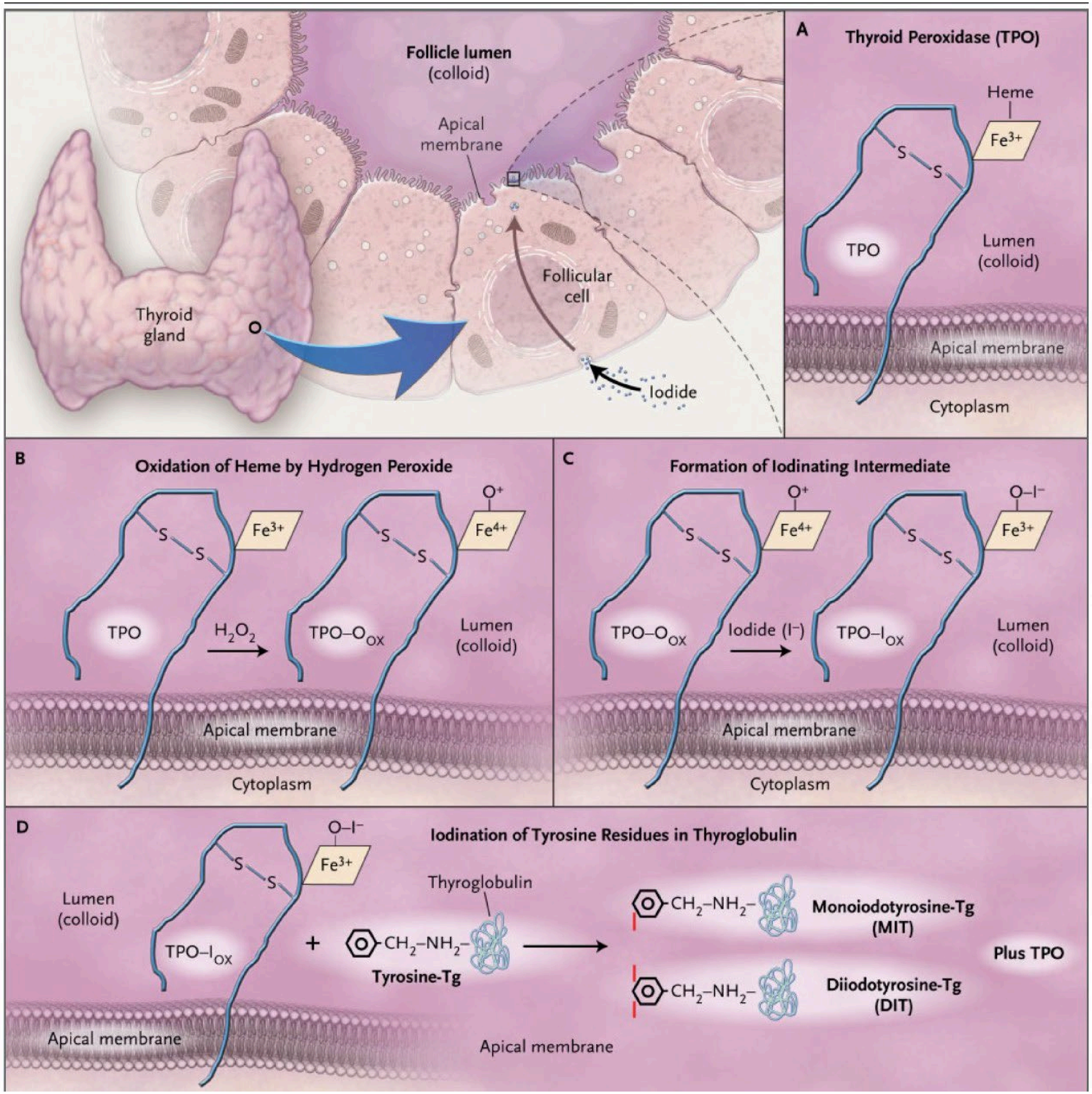
# Physiology of Thyroid Gland

Thyrotrophin releasing hormone




Thyroid stimulating hormone

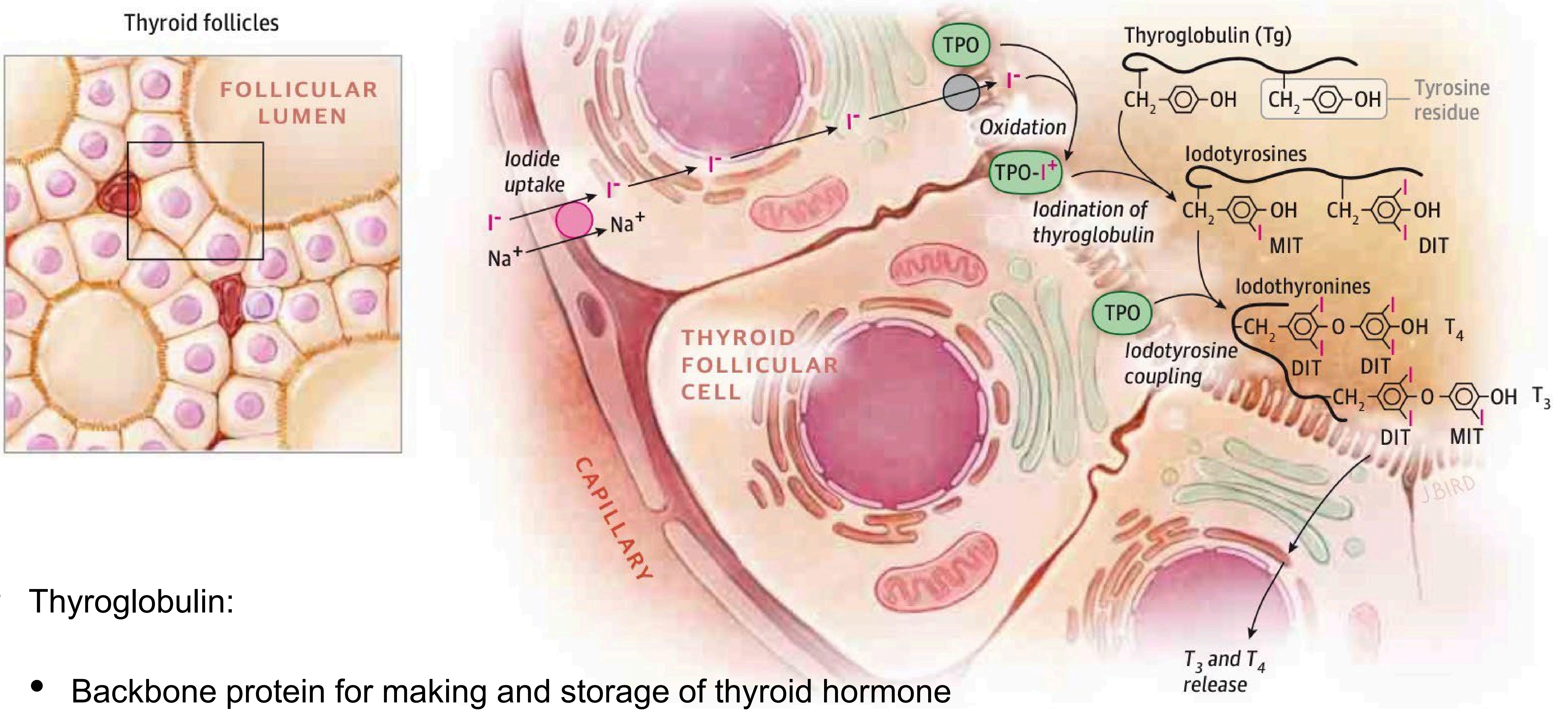
**Circulating levels (>99% protein-bound)**




1) Iodide uptake by thyroid follicular cells
2) Oxidation of iodide by thyroid peroxidase (TPO)
3) TPO catalyzes the subsequent steps of thyroid hormone synthesis
4) Iodination of tyrosine residues in thyroglobulin to form iodotyrosines
5) Coupling of 2 iodotyrosines to form the iodothyronines bound to thyroglobulin
6) The iodothyronines are stored in the follicular lumen as colloid and ingested by thyroid follicular cells
7) Within the follicular cells, thyroid hormones thyroxine (T4) and triiodothyronine (T3) are released from thyroglobulin and enter the bloodstream

Cooper D. N Eng J Med 2005

A Biosynthesis of thyroid hormone




- Thyroglobulin:
  - Backbone protein for making and storage of thyroid hormone
- Thyroperoxidase (Previously known as thyroid microsomal enzyme):
  - An enzyme for organification process and synthesis of thyroid hormone
- The thyroid gland produces predominantly the prohormone T4 together with a small amount of the bioactive hormone T3
  - Most T3 is produced by enzymatic outer ring deiodination (ORD) of T4 in peripheral tissues

Burch H et al. JAMA 2015

# Mechanism of Thyroxine Actions

**T3**

**Avid cellular uptake**

**Intracellular protein receptor (T3 > T4)**

**Modulation of nuclear RNA**

**Protein Synthesis**

**Enzyme Synthesis**

**Mitochrondrial Oxidative Phosphorylation**

# Tests of Thyroid Function

- Serum **Total** T4 levels
  - A measurement of total T4 which is bound to plasma binding proteins
  - 99.96% bound to thyroid hormone-binding globulin (TBG) and thyroxine-binding prealbumin (TBPA)
  - **Affected by disease states which alter TBG level** (e.g. increase in pregnancy and decrease in conditions with hypoalbuminaemia)

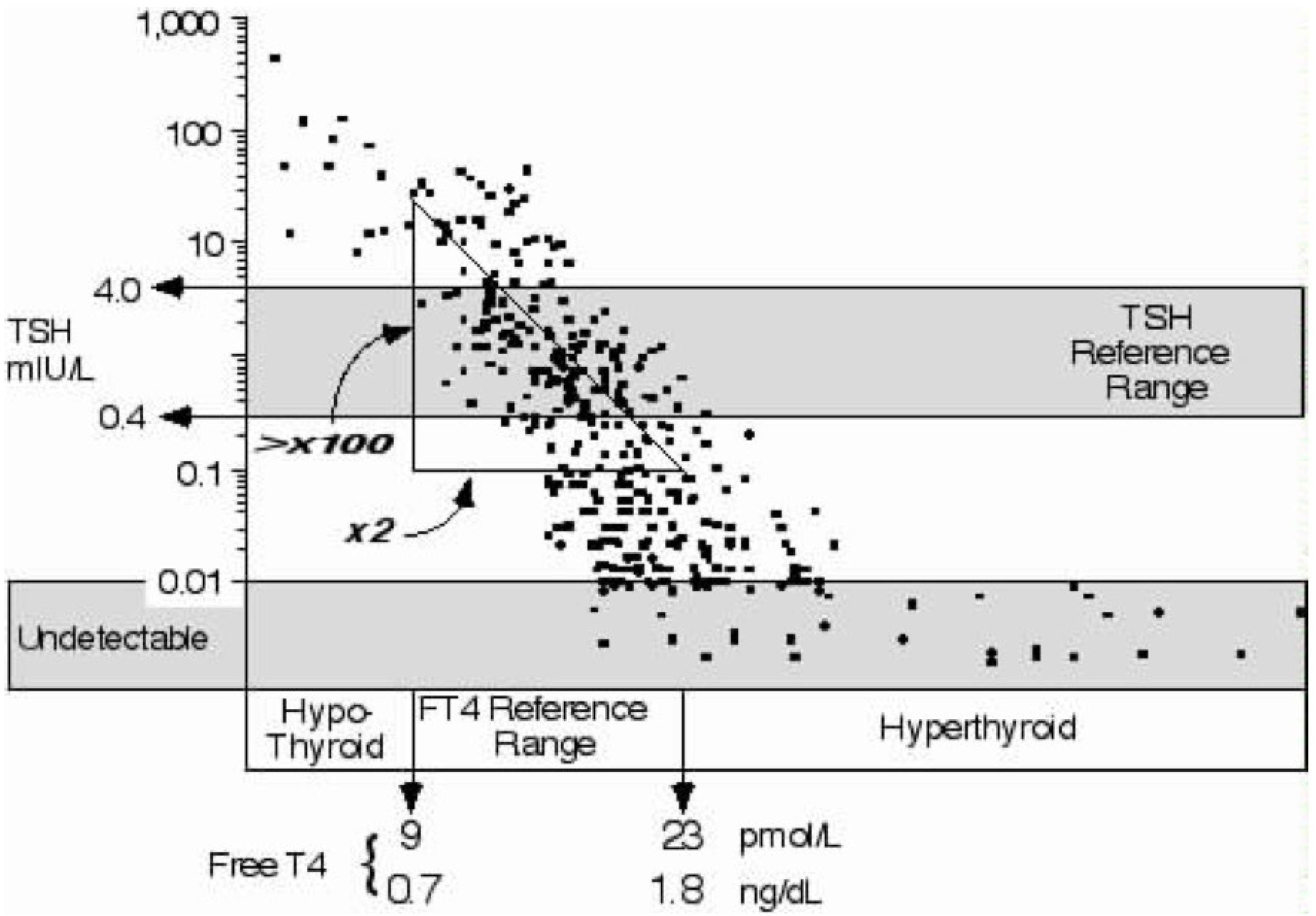
# Tests of Thyroid Function

- Serum **free** thyroid hormones (fT4 and fT3) levels
  - Measurements of unbound free circulating hormone levels
  - 0.02% of T4 and 0.2% T3 exist as free forms
- Serum TSH
  - Suppressed in thyrotoxicosis
  - Low or inappropriately normal in central or secondary hypothyroidism (hypothalamic pituitary failure)
  - Elevated in primary hypothyroidism

# Screening for Thyroid Dysfunction

**TSH is used to screen for *primary* thyroid dysfunction**

- Two reasons for using a TSH-centered strategy for ambulatory patients:
  - **Inverse log/linear relationship**
    - Small alterations in fT4 lead to a large response in serum TSH
  - In early stages of thyroid dysfunction, **serum TSH abnormality will precede the development of abnormal fT4**
    - TSH responds exponentially to even subtle fT4 changes within the population reference limits



# Screening for Thyroid Dysfunction

**TSH is used to screen for *primary* thyroid dysfunction**

- Prerequisites
  - **Intact hypothalamic-pituitary axis**
  - Thyroid status is stable (i.e. no recent anti-thyroid therapy for hyperthyroidism)

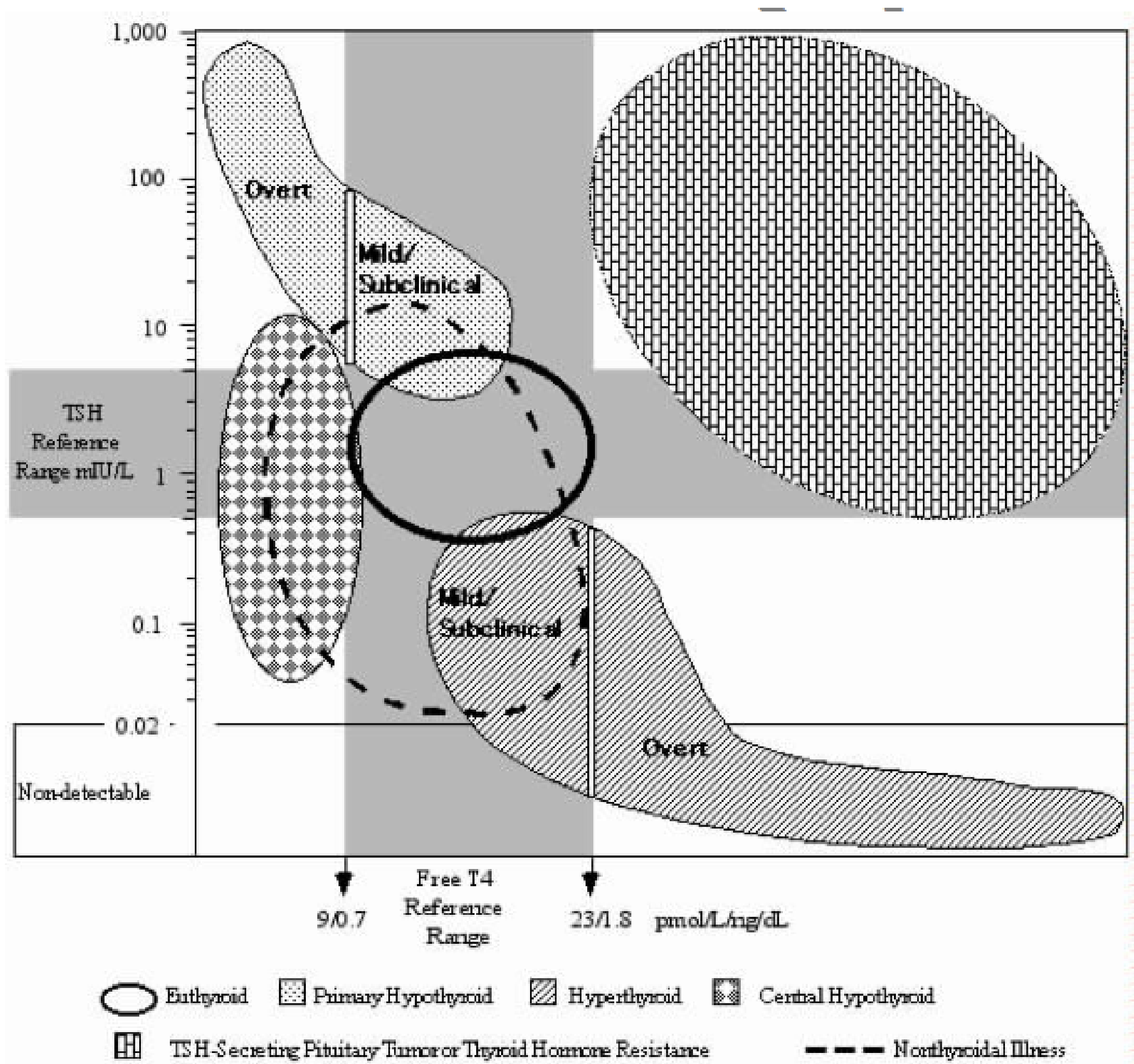
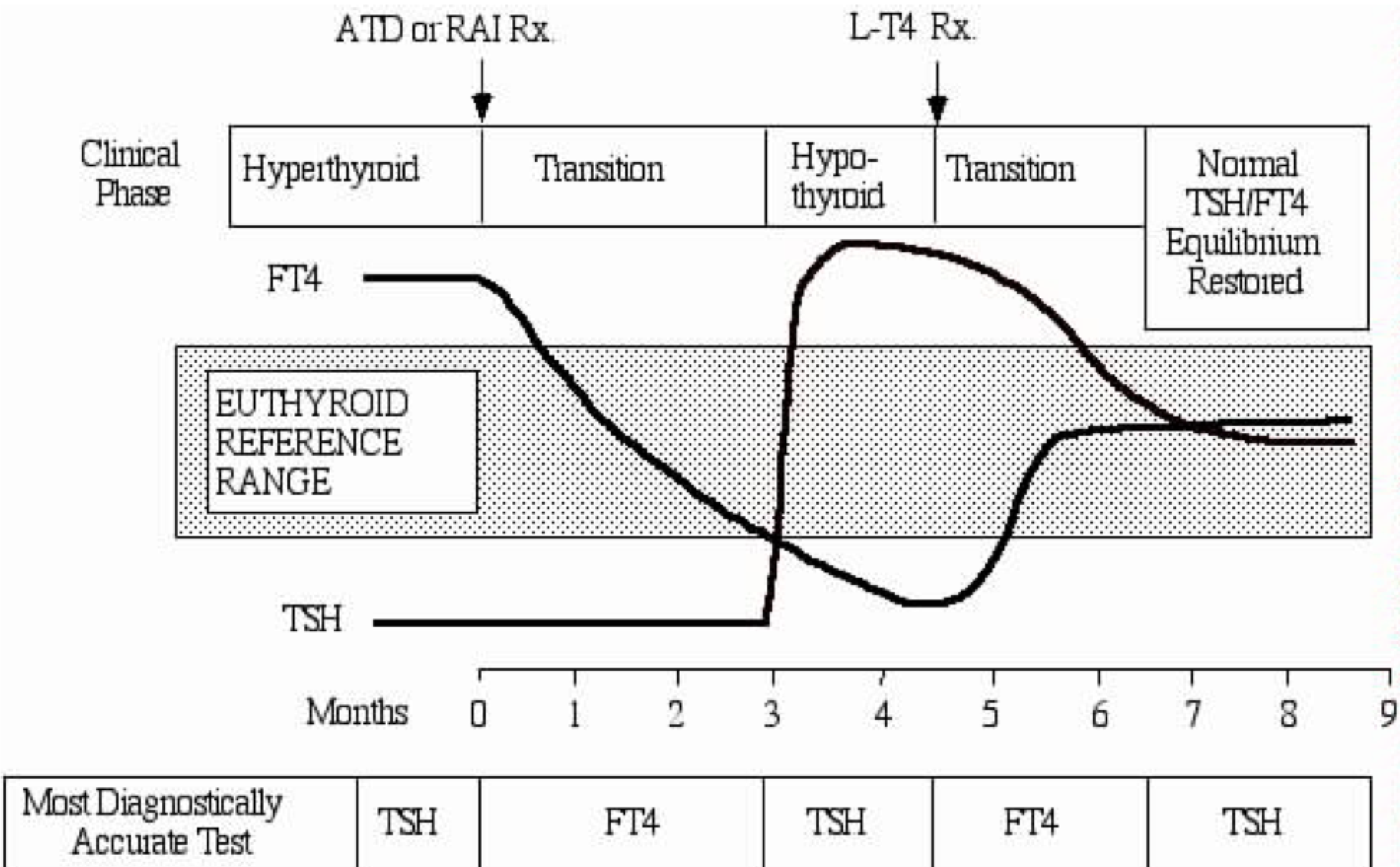



*Fig 4. The serum TSH/FT4 relationship typical of different clinical conditions*

Laurence M Demers et al.
NACB: Laboratory Support for the Diagnosis and Monitoring of Thyroid Diseases

The National Academy of Clinical Biochemistry

LABORATORY SUPPORT FOR THE DIAGNOSIS OF THYROID DISEASE



# Hyperthyroidism

# Symptoms of Hyperthyroidism

- Weight loss
- Palpitations
- Nervousness
- Easy fatiguability
- Excessive sweating
- Heat intolerance
- Hyperkinaesia
- Increased bowel motions
- Hair loss
- Visual complaints

# Signs of Hyperthyroidism

- Goitre
- Tachycardia
- Sweaty palms
- Hand tremor
- Pretibial myxoedema

- Periorbital edema
- Lid lag
- Lid retraction
- Exophthalmos (proptosis)
- Ophthalmoplegia
- Corneal involvement
- Sight loss

# Causes of Thyrotoxicosis

- Graves' disease
- Toxic multi nodular goitre
- Toxic solitary adenoma
- Thyroiditis (subacute / painless / medications including immune checkpoint inhibitors)
- TSH secreting pituitary adenoma
- Molar hyperthyroidism
- Factitious causes

# Graves' disease

- Autoimmune disease
  - Lymphocytes acting against self-antigen (i.e TSH receptor)
- Thyrotropin-receptor antibody (TRAb) stimulates TSH receptor on the thyroid gland
  - Stimulates thyroid cell hyperplasia and increases thyroid hormone release resulting in hyperthyroidism, which in turn suppresses TSH release for the pituitary gland

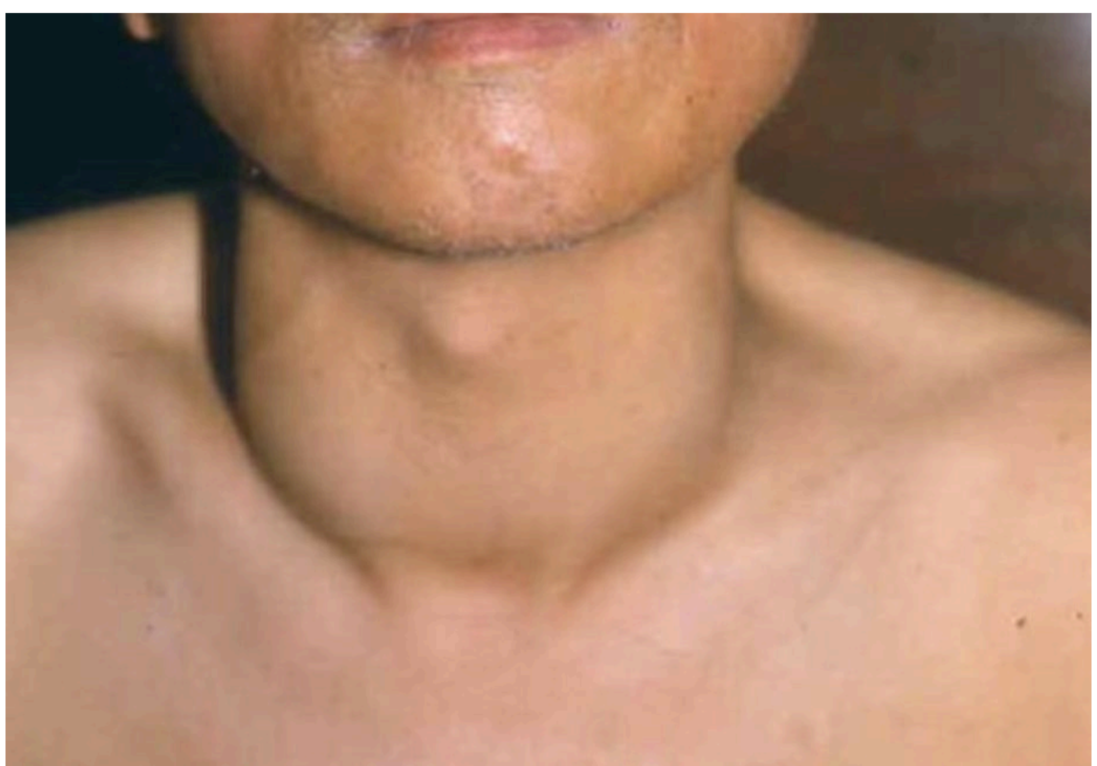
# Graves' disease

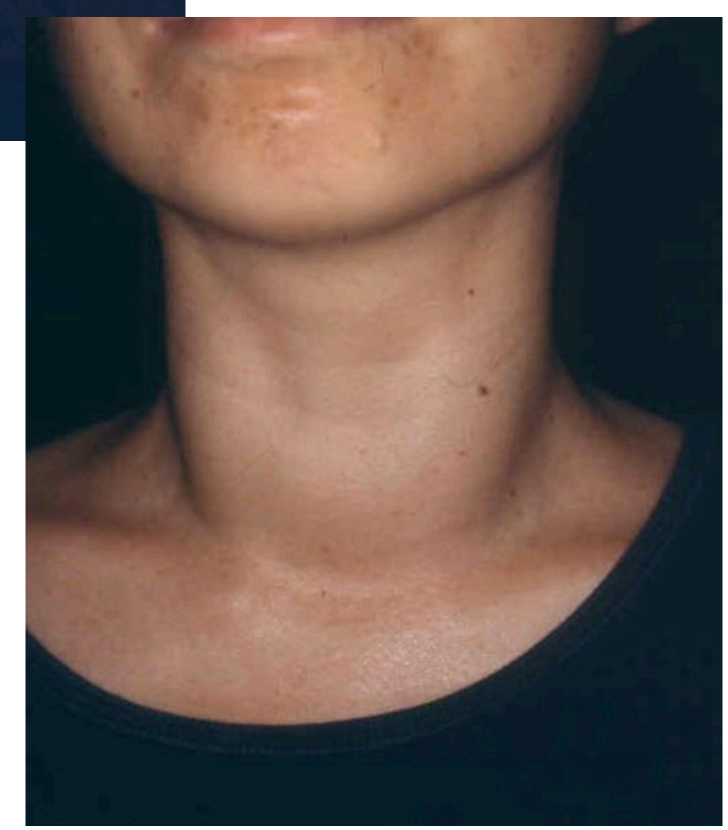
- Clinical Diagnosis!
- Diffusely enlarged thyroid gland (i.e. not nodular)
- Raised fT4 / fT3 and suppressed TSH level
- Positive anti-TSHR antibody
- Diffusely increased uptake on radioiodine scintillation scan
- Diffusely increased vascularity of the thyroid gland on ultrasound

# Clinical Presentation of Graves' disease

- Female preponderance - F:M 4.8:1
- Highest incidence 20-50 years
- Diffuse goitre with thyroid bruit
- Symptoms and signs of hyperthyroidism
- Graves' ophthalmopathy
- Pretibial myxoedema
- Related autoimmune diseases:
  - Myasthenia gravis, type 1 diabetes mellitus
- Complications:
  - Atrial fibrillation, heart failure, thyroid storm, thyrotoxic periodic paralysis

# Clinical Presentation of Graves' disease





# Clinical Presentation of Graves' disease

# TRAB or Anti-TSHR Ab in Graves' Disease

- Prognostic indicator of the outcome of anti-thyroid drug treatment
  - Positive TRAb at the end of a period of therapy indicates higher chance of relapse
  - Negative TRAb more likely to have a prolonged remission
- Forecasting neonatal Graves' disease
  - High level of TRAb associated with neonatal Graves' disease due to transplacental passage of antibody
- Aids in establishing the diagnosis of Graves' disease
  - Usually not necessary as nearly 100% of patients with active Graves' disease are positive for TRAb
  - Level of antibody decreases with anti-thyroid dug treatment

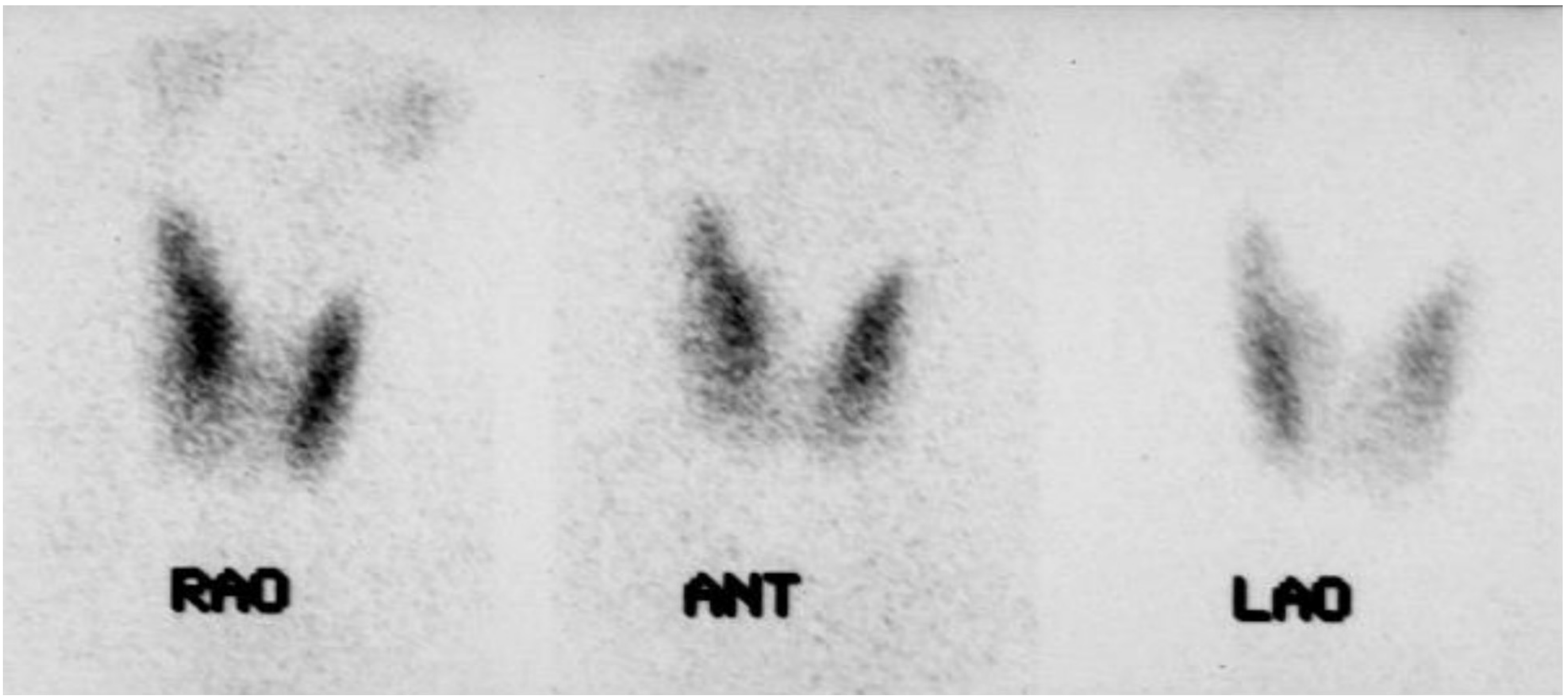
# Thyroid Imaging

- Ultrasound
  - Useful especially in the context of palpable thyroid nodules to evaluate for the presence of suspicious sonographic features
- Thyroid scan
  - Suggested when thyroid nodularity co-exists with hyperthyroidism
  - If thyroiditis is suspected

# Thyroid Scan

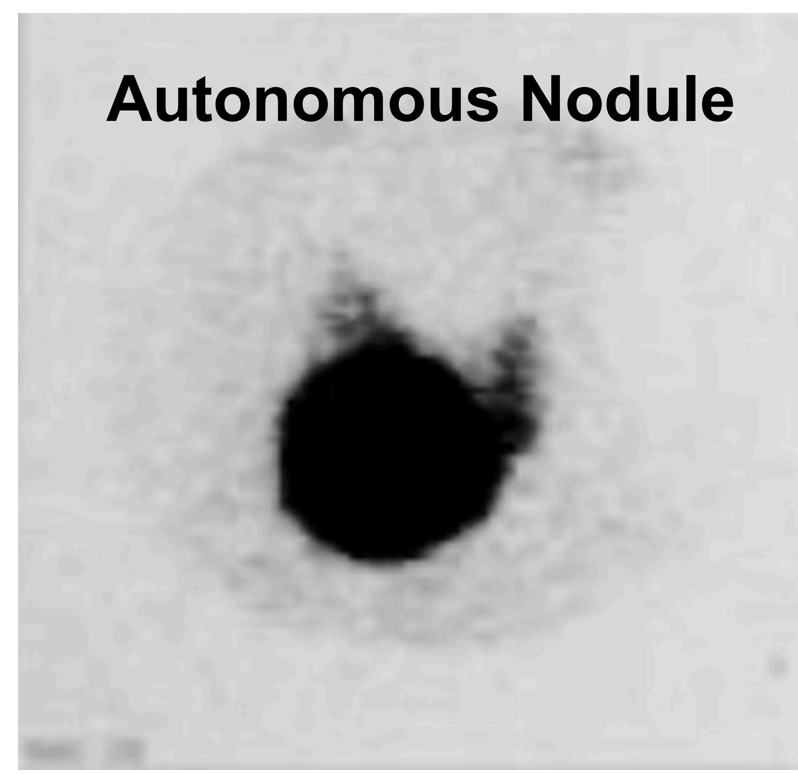
- Conditions with increased uptake on thyroid scan
  - Graves' disease - diffuse uptake
  - Multinodular goitre - heterogeneous uptake
  - Toxic adenoma - focal area of increased uptake with reduced uptake in the rest of the gland
  - TSH-secreting pituitary adenoma - diffuse uptake
- Conditions with decreased uptake on thyroid scan
  - Thyroiditis
  - T4 overdose
  - Iatrogenic

# Thyroid Scan




**Normal Thyroid**

**Graves' disease**



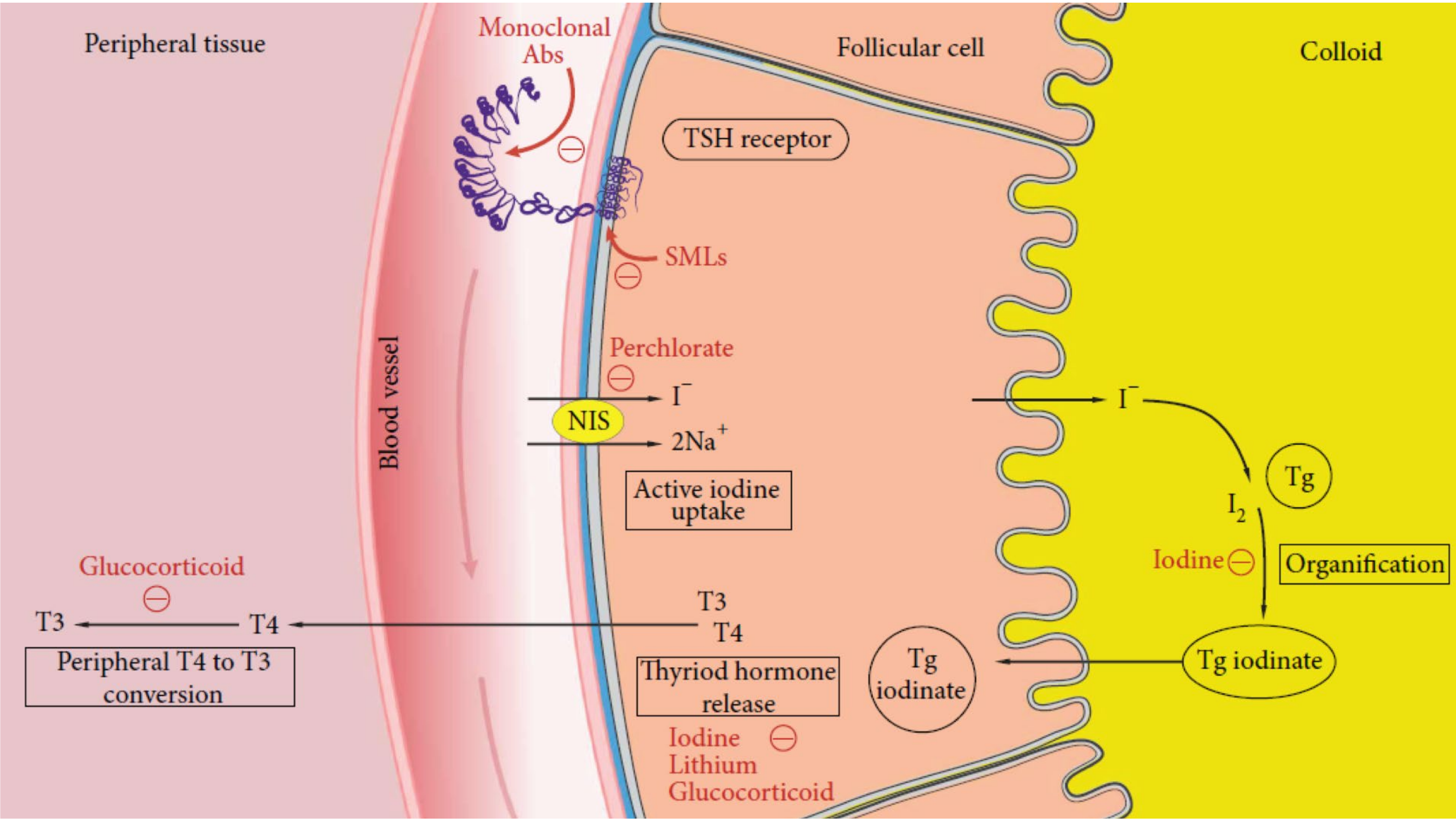
**Autonomous Nodule**

# Management of Hyperthyroidism

- Anti-thyroid drugs
  - Thiouracil derivatives
    - Methimazole, carbimazole (prodrug), prophylthiouracil
  - Lithium
- Surgery
- Radioactive iodine therapy
- Ancillary drug therapy
  - Beta-blockers
  - Iodide (e.g. Lugol's iodine)
  - Sedatives

# Anti-thyroid drugs

- Thiouracil derivatives
  - Methimazole (MMZ), carbimazole (CMZ)
  - Prophylthiouracil (PTU)
    - Shorter half-life
  - 12-18 months of therapy
  - Changes of relapse >60%
  - Sometimes block and replacement therapy (Thiouracil + T4)
- Lithium
  - Block Na-I transporter
  - For patients with intolerance or contraindications to MMZ/CMZ/PTU




Suwansaksri N. et al, Int J Endocrinol 2018

# Mechanisms of Actions of Thiouacil Derivatives (MMZ / CMZ / PTU)

- Inhibit thyroid hormone synthesis
  - Inhibit organification of iodide
  - Inhibit coupling of iodotyrosines
- Inhibit T4 to T3 conversion (PTU)
- Immunosuppressive effects on thyoid autoantibodies
  - Action on intrathyoidal antigen-presenting cells

# Pharmacological Properties of MMZ / CMZ / PTU

- Unrestricted absorption
- Actively concentrated by the thyroid within minutes
- Long T1/2, especially MMZ (can be given as daily dose)
- PTU cross placenta relatively less than MMZ / CMZ
- Both transferrable to milk, MMZ/CMZ > PTU
- Teratogenicity
- MMZ / CMZ more severe than PTU
  - Aplasia cutis, choanal and esophageal atresia with MMZ / CMZ
  - Preauricular sinus/ fistula, urinary tract abnormalities with PTU
- Preconception counselling and discussions of fertility wish
  - Change MMZ / CMZ to PTU if planning for conception and remain until end of first trimester
  - Change back to MMZ / CMZ in second trimester for less hepatotoxicity

# Side Effects of Anti-Thyroid Drugs

- Rash (5%)
- Agranulocytosis (0.1%)
  - Idiosyncratic reactions
  - Reversible with discontinuation of ATD and GCSF
  - Usually in the first 2-3 months
  - Increased with age (>40 yo)
  - With high doses
  - Usual presenting symptoms: fever, sore throat
- Rare
  - Cholestatic jaundice, hepatotoxicity, acute arthralgia, ANCA induced vasculitis

# Predictors of Relapses

- Larger goitre
- Positive family history
- Higher fT3 or fT3/fT4 ratio on presentation
- Shorter course of treatment
- Failure to normalize TSH during treatment
- High anti-TSHR levels

# Indications of Surgical Treatment (Total Thyroidectomy)

- Young age
- Failed medical treatment
- Intolerance / contraindications to anti-thyroid medications
- Refused radioactive iodine
- Significant Graves' ophthalmopathy
- Expecting pregnancy
- Large goitre especially with compressive effects

# Preoperative optimization

- Anti-thyroid medications till euthyroid
- Beta-blockers for at least 2 weeks
- (Lugol’s iodine)

# Complications of Thyroid Surgery

- Primary hypothyroidism requiring lifelong thyroxine (T4) replacement
- Vocal cord dysfunction (Transient vs permanent)
  - Recurent laryngeal nerve
  - External branch of the superior larngyngeal nerve
- Hypoparathyroidism (Transient vs permanent)
- Bleeding
- Tracheomalacia
- Wound complications
  - Seroma, hypertrophic scar
- Precipitation of thyroid storm
- Risks related to general anaesthesia

# Radioactive Iodine

- In use since late 1940s
- Advantages
  - Low relapse rate (15% after 1 dose)
  - Simple and economical
  - No immediate complications
- Indications:
  - Relapsed thyrotoxicosis especially after previous thyroid surgery (subtotal thyroidectomy / hemithyroidectomy)
  - Thyrotoxic heart disease
  - Thyrotoxic periodic paralysis
  - Contraindications to surgery
  - Toxic multi nodular goitre

# Radioactive Iodine

- Alpha emission of $^{131}I$ mostly not absorbed
- Beta emission almost completely absorbed
- Actions
  - Necrosis of follicular cells, fibrosis and disappearance of colloid
  - Non-destroyed cells have limited replication
  - Effects on thyroid autoimmunity
    - ?Destruction of radiosensitive intra-thyroid T suppression cells
    - ?Release of thyroid antigens stimulating antiTSHR Ab

# Complications and Precautions of Radioactive Iodine Therapy

- Primary Hypothyroidism
  - Transient 3.5 - 28%
  - Permanent 10-15% in the first 2 years, then 3% per year
- Fetal risk (Definite)
  - Transient radiation induced changes in gametes
    - Avoid pregnancy for 6-12 months
  - Crosses placenta and concentrated by fetal thyroid (>12 weeks)
  - Excreted in milk
    - Therefore contraindicated in pregnant and lactating women
- NO increase in thyroid carcinoma / leukaemia / transmissible genetic damage
- Increased risk of de novo Graves' ophthalmopathy development, and/or exacerbation of pre-existing mild Graves' ophthalmopathy after RAI
- Precipitate thyroid storm due to radiation thyroiditis

# Summary for modalities of treatment

| *Clinical situations* | *RAI* | *ATD* | *Surgery* |
|---|---|---|---|
| Pregnancy[a] | x | √√ / ! | √ / ! |
| Advanced age, comorbidities with increased surgical risk and/or limited life expectancy | √√ | √ | x |
| Patients with previously operated or externally irradiated necks | √√ | √ | ! |
| Lack of access to a high-volume thyroid surgeon | √√ | √ | ! |
| Symptoms or signs of compression within the neck | √ | - | √√ |
| Thyroid malignancy confirmed or suspected | x | - | √√ |
| Large goiter/nodule | √ | - | √√ |
| Goiter/nodule with substernal or retrosternal extension | √ | - | √√ |
| Coexisting hyperparathyroidism requiring surgery | - | - | √√ |

√√ = preferred therapy; √ = acceptable therapy; ! = cautious use; - = not usually first line therapy but may be acceptable depending on the clinical circumstances; X = contraindication.
[a]For women considering a pregnancy within 6 months, see discussion in Section [T2].

ATA guidelines; Thyroid 2015

# Graves' ophthalmopathy (GO)

- Dysthyroid eye disease / Thyroid eye disease
- Pathogenesis:
  - Orbital fibroblasts being the target cells
  - Ability to differentiate into adipocytes in a subpopulation
  - Increased TSH receptor expression on orbital fibroblast / adipocytes as a consequence of adipocyte differentiation
  - TSH also stimulates orbital adipogenesis in GO but not normal orbital preadipocytes
  - TSHR Ab stimulates orbital adipocytes
  - TSHR Ab level correlates with clinical scores of GO

# Graves' ophthalmopathy (GO)

- Histological changes
  - Extraocular muscles
    - Edema, mononuclear cell infiltration, mucopolysaccharides accumulation and fibrosis
- Retrobulbar fat
    - Lymphocytic infiltration, fibrous tissue formation, accumulation of collagen tissues with hyaluronic acid
- Optic nerve
    - Atrophy, replaced by fibrous and fatty connective tissues

# Graves' ophthalmopathy (GO)

- Patients can be hyper-, hypo- or euthyroid
- 80% of patients with GO first develop eye signs within 18 months of the diagnosis of thyrotoxicosis
- Debatable whether mild GO should affect choice of treatment of hyperthyroidism
  - Needs steroid cover post RAI especially with risk factors for GO exacerbation (e.g. chronic smokers, pre-existing mild GO, high anti-TSHR Ab)
- Those with significant GO (active, moderate to severe GO, dysthyroid optic neuropathy)
  - Not for RAI as definitive therapy
  - May consider total thyroidectomy to prevent relapses of thyrotoxicosis and destabilization of eye conditions

# Classification of GO

**Table 3** Classification of severity of Graves' orbitopathy (GO).

| Classification | Features |
|---|---|
| Mild GO | Patients whose features of GO have only a minor impact on daily life that have insufficient impact to justify immunomodulation or surgical treatment. They usually have one or more of the following:<br>• minor lid retraction (<2 mm)<br>• mild soft-tissue involvement<br>• exophthalmos<br>• <3 mm above normal for race and gender<br>• no or intermittent diplopia and corneal exposure responsive to lubricants |
| Moderate-to-severe GO | Patients without sight-threatening GO whose eye disease has sufficient impact on daily life to justify the risks of immunosuppression (if active) or surgical intervention (if inactive). They usually have two or more of the following:<br>• lid retraction ≥ 2 mm<br>• moderate or severe soft-tissue involvement<br>• exophthalmos ≥ 3 mm above normal for race and gender<br>• inconstant or constant diplopia |
| Sight-threatening (very severe) GO | Patients with dysthyroid optic neuropathy and/or corneal breakdown |

Bartalena L et al. EUGOGO 2021 Guidelines, Eur J Endocrinol 2021

# Assessment of Activity and Severity of GO

**Table 2** Assessment of activity by the clinical activity score (CAS)*. CAS < 3 = inactive GO; CAS ≥ 3 = active GO. A ten-item CAS, including an increase in exophthalmos of ≥2 mm, a decrease in eye motility of ≥8° or a decrease in visual acuity in the last 1–3 months, is useful to assess progression of GO after the first visit.

| **Assessment of activity** |
|---|
| 1. Spontaneous retrobulbar pain |
| 2. Pain on attempted upward or downward gaze |
| 3. Redness of eyelids |
| 4. Redness of conjunctiva |
| 5. Swelling of caruncle or plica |
| 6. Swelling of eyelids |
| 7. Swelling of conjunctiva (chemosis) |

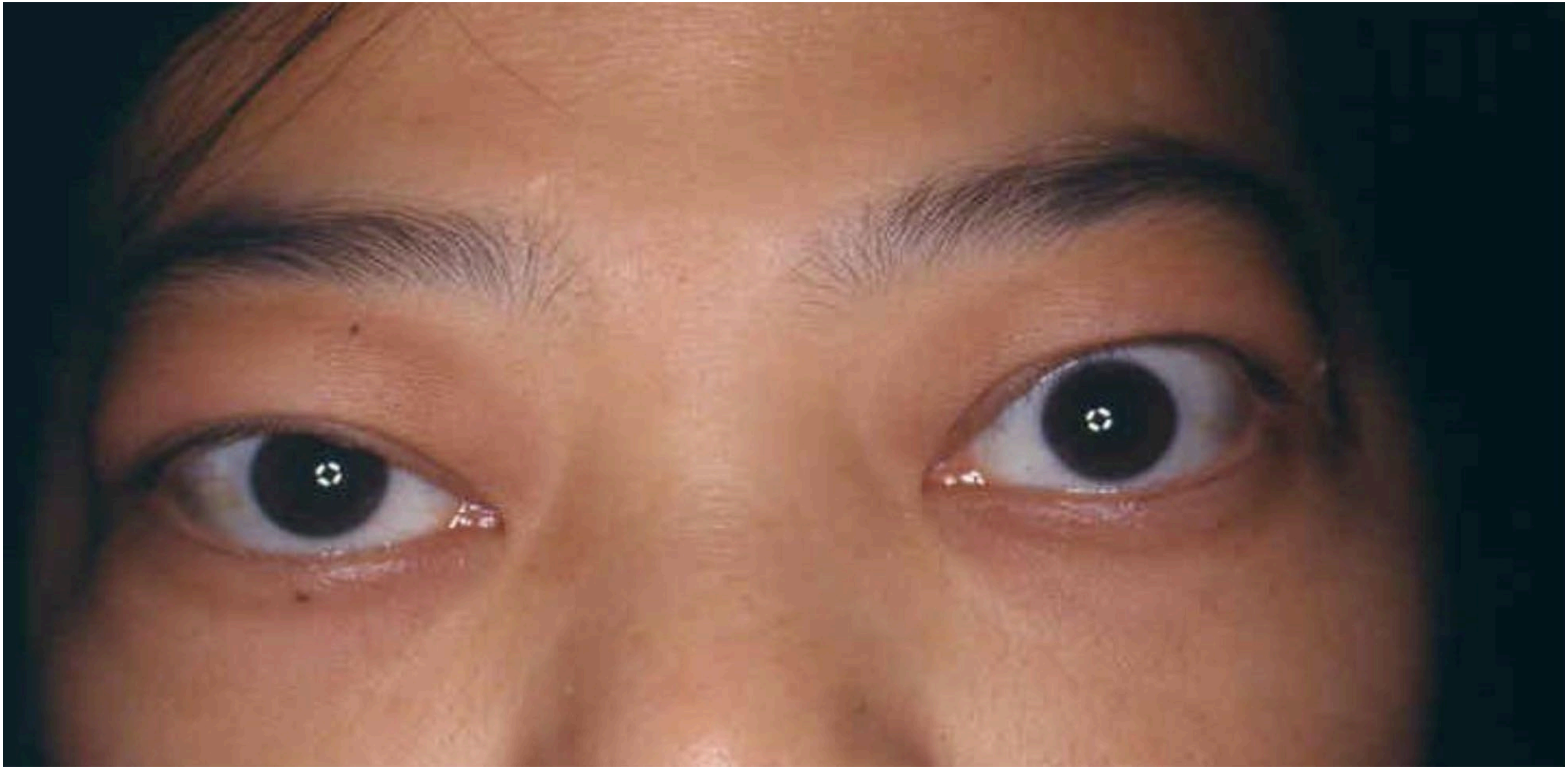
*Modified according to Wiersinga *et al.* (14) and reproduced with permission.

Bartalena L et al. EUGOGO 2021 Guidelines, Eur J Endocrinol 2021

# Assessment of Activity and Severity of GO

- Comprehensive ophthalmological assessment
  - Visual acuity, intraocular pressure, colour vision, optic nerve functions, extra ocular movements
- Radiological assessment
  - MRI orbit (more preferable) or plain CT orbit
  - Crowding of the orbital apices, soft tissue changes, enlargement of extra-ocular muscles, optic nerve calibre

# Mild GO



# Management of mild GO

**MILD GO**

**General recommendations**

- Refrain from smoking
- Treat thyroid dysfunction (preferably with antithyroid drugs, especially if risk factors for deterioration/progression of GO are present (see below)
- Avoid iatrogenic hypothyroidism in treating patients with GD/GO
- Referral to thyroid-eye clinics if risk factors present (active GO, smoker, high TSHR-Ab, unstable / severe hyperthyroidism)
- Search for dry eye syndrome

**Management**

Local treatment
- Artificial tears, especially when dry eye present
- Ophthalmic gels (cornea protection during the night)

Systemic adjunct therapy for active GO
- Selenium supplementation for six months (fasting intake)

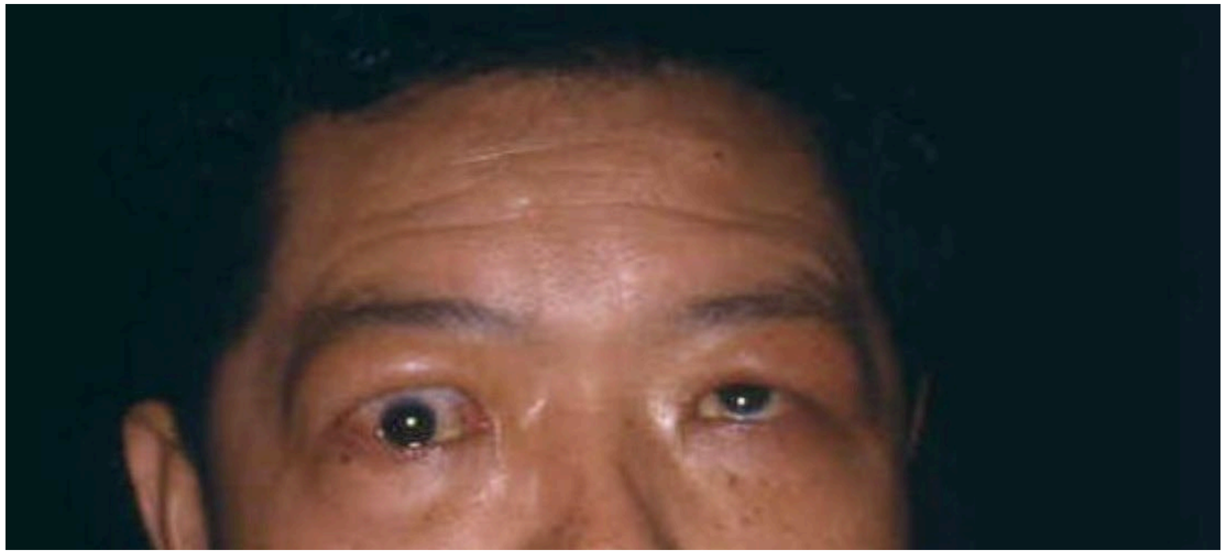
Quality of life markedly impaired

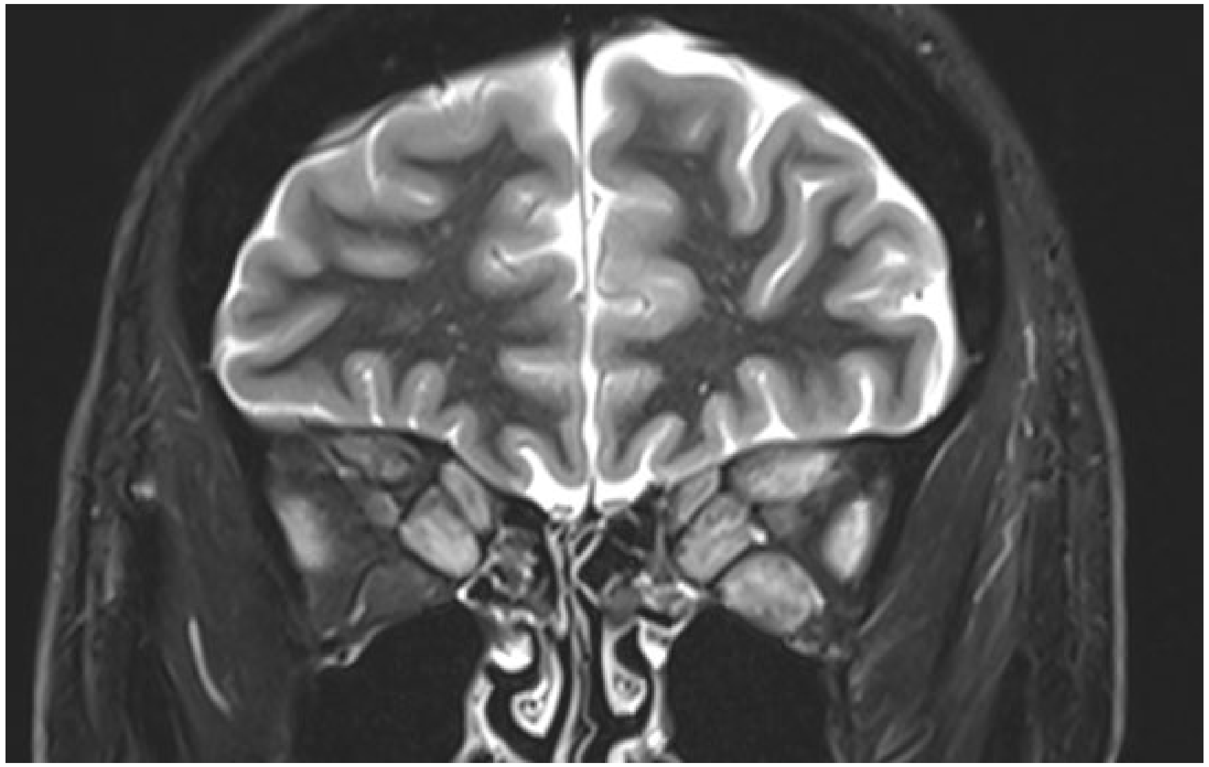
↓

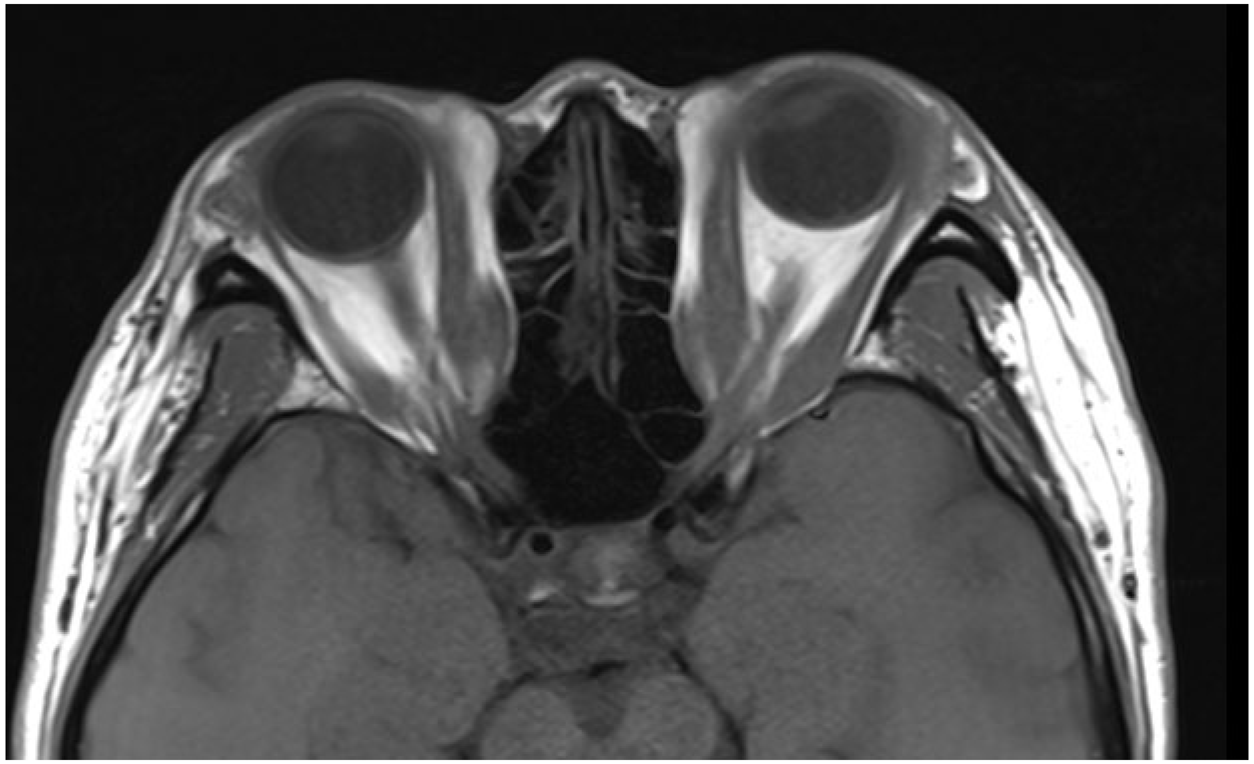
Discuss low dose immunomodulatory (active GO) or rehabilitative surgery (inactive GO) following extensive counseling and shared decision

Bartalena L et al. EUGOGO 2021 Guidelines, Eur J Endocrinol 2021

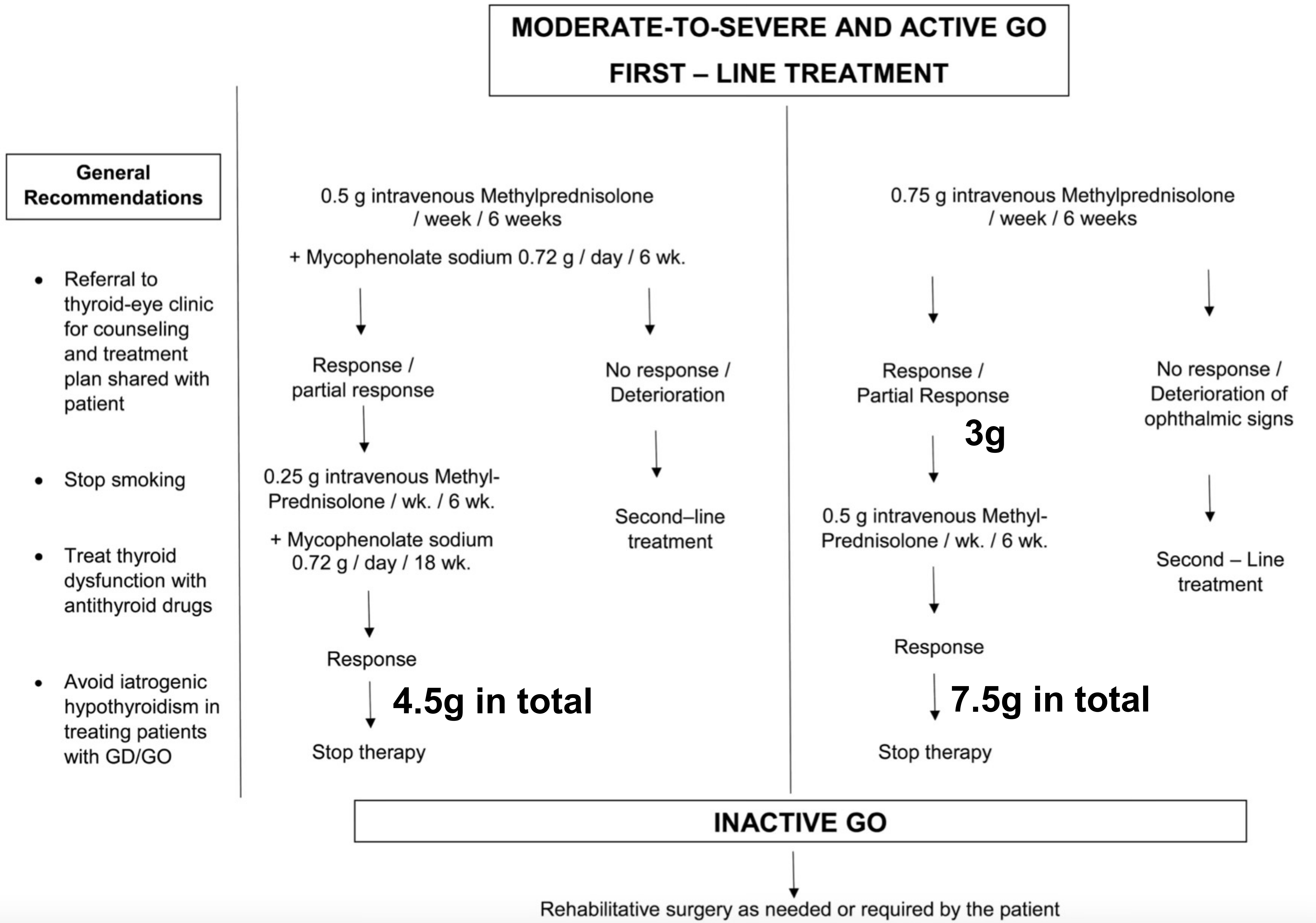
# More Severe GO







# Management of moderate to severe active GO

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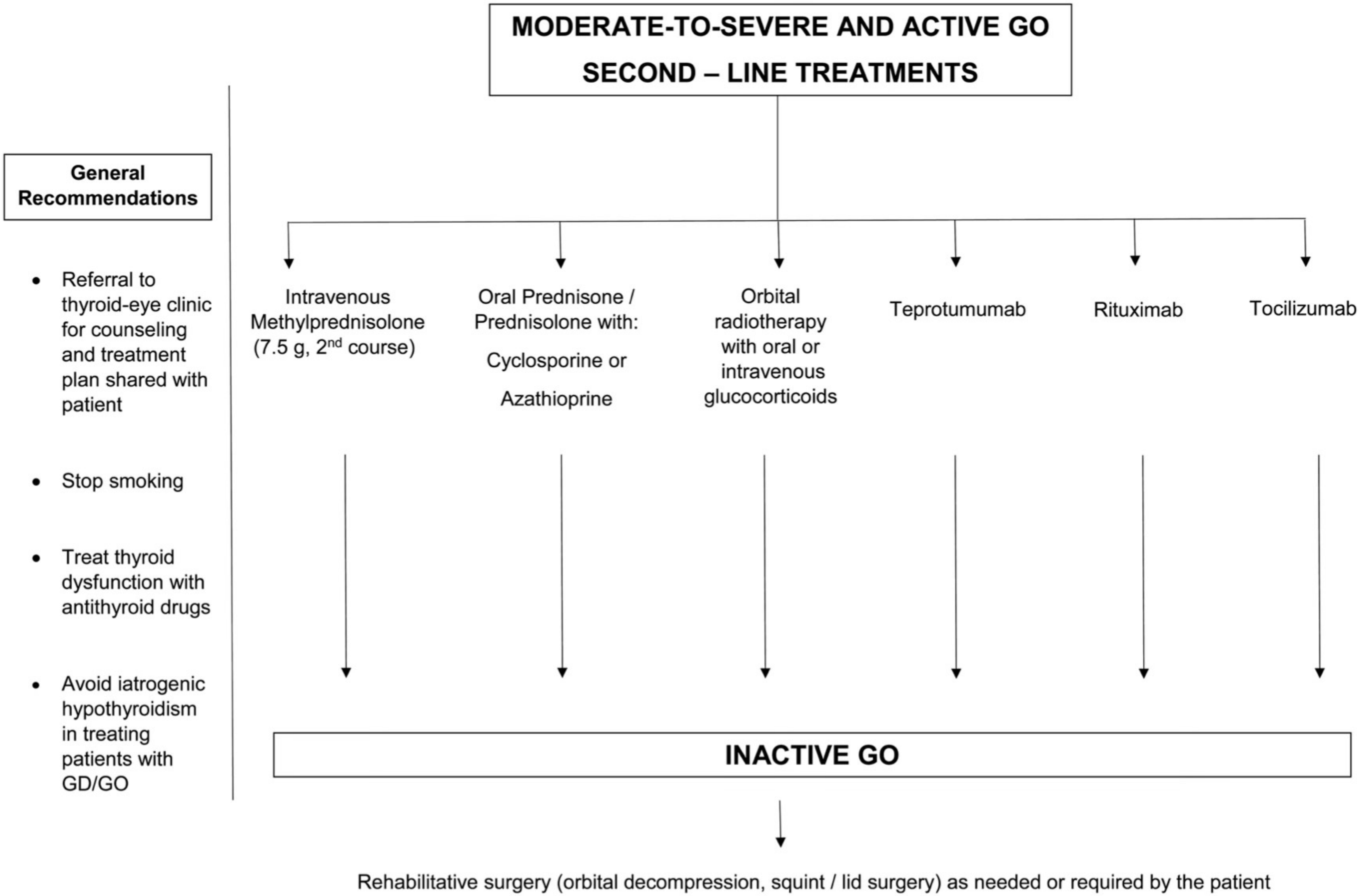
# Management of moderate to severe active GO




Bartalena L et al. EUGOGO 2021 Guidelines, Eur J Endocrinol 2021

# Management of sight-threatening GO

**SIGHT - THREATENING GO (Optic Neuropathy)**

**General recommendations**

- Immediate referral to thyroid-eye clinic
- Stop smoking
- Avoid radioactive iodine treatment
- Stabilize thyroid dysfunction with antithyroid drugs
- Avoid iatrogenic hypothyroidism in treating patients with GD/GO

**Specific Management**

Intravenous methylprednisolone (0.5 – 1 gram, as single dose repeated on three consecutive or alternate days)

↓

Daily monitoring of ophthalmic parameters

↓

After one week, evaluation if therapy can be continued

↓ Yes

↓ Further with intravenous methylprednisolone as in week one

- ↓ Response → 0.5 g intravenous methylprednisolone 1x / wk. (cumulative dose <8 g / cycle)
- ↓ Partial response → Urgent orbital decompression surgery (Imaging recommended)

↓ No response and/or deterioration of ophthalmic signs → Urgent orbital decompression surgery (Imaging recommended)

Bartalena L et al. EUGOGO 2021 Guidelines, Eur J Endocrinol 2021

# Thyrotoxic Periodic Paralysis (TPP)

- Mainly in Orientals, rare in Caucasians
- Predominantly in male patients
  - M:F - 25% vs 0.8%
- Involves skeletal muscles
  - Mainly motor involvement; seldom respiratory muscles
- Hypokalaemia
  - Risk of cardiac arrhythmia
- Occurs during hyperthyroidism, not when euthyroid
- Spontaneous recovery, which is accelerated by IV potassium infusion
- Attacks prevented by low salt diet, appropriate carbohydrate intake, spironolactone, propranolol
- Regular potassium supplements not necessary; only given when symptomatic

# Pathogenesis of TPP

- Hypokalaemia due to shifting of potassium from extracellular to intracellular compartments
- Enhanced $Na^+/K^+$ - ATPase activity especially after a high carbohydrate load
- Enhanced insulin response
- Insulin drives $K^+$ into cells

# Hypothyroidism

# Case Vignette

- Ms Chan
- F/30, Non smoker
- Good past health
- Family hx of thyroid disease
- Malaise, feeling cold, constipation and weight gain
- Central neck swelling
- Blood tests revealed low thyroid hormone and high TSH levels
- ECG showed sinus bradycardia

# Symptoms of Hypothyroidism

**Table 1. Common Symptoms of Hypothyroidism**

| | | |
|---|---|---|
| Arthralgias | Dry skin | Menorrhagia |
| Cold intolerance* | Fatigue* | Myalgias |
| Constipation | Hair thinning/ hair loss | Weakness |
| Depression | Memory impairment | Weight gain |
| Difficulty concentrating | | |

*—*Most common.*

Am Fam Physician. 2012;86(3):244-251

# Signs of Hypothyroidism

**Table 2. Clinical Signs of Hypothyroidism**

| | |
|---|---|
| Bradycardia | Laboratory results *(continued)* |
| Coarse facies | Increased creatine kinase |
| Cognitive impairment | Increased low-density lipoprotein cholesterol |
| Delayed relaxation phase of deep tendon reflexes | Increased triglycerides |
| Diastolic hypertension | Normocytic anemia |
| Edema | Proteinuria |
| Goiter | Lateral eyebrow thinning |
| Hypothermia | Low-voltage electrocardiography |
| Laboratory results | Macroglossia |
| Elevated C-reactive protein | Periorbital edema |
| Hyperprolactinemia | Pleural and pericardial effusion |
| Hyponatremia | |

Am Fam Physician. 2012;86(3):244-251

# Delayed reflexes in Hypothyroidism

# Causes of Hypothyroidism

- Primary hypothyroidism
  - Autoimmune
    - Atrophic thyroiditis
    - Hashimoto’s thyroiditis
  - Radioactive iodine therapy
  - Subtotal thyroidectomy
  - Subacute thyroiditis
  - Medications (e.g immune check-point inhibitors)
  - Congenital hypothyroidism
  - Excessive iodide intake
  - Iodine deficiency
    - Borderline iodine intake in Hong Kong
- Central / secondary hypothyroidism

# Painless Thyroiditis

- Lymphocytic thyroiditis
- Atrophic thyroiditis
- Hashimoto's thyroiditis
  - Enlarged thyroid with lymphocytes
- Postpartum thyroiditis
- Variable prognosis depending on residual thyroid reserve

# Hashimoto's Thyroiditis

- Autoimmune disease
- Common cause of primary hypothyroidism
- Low T4 and high TSH
- Positive anti-thyroid peroxidase antibody
- Enlarged goitre due to increased fibrosis and lymphocytic infiltration
- Common in middle-aged women
- May be associated with other autoimmune diseases

# Subacute Thyroiditis

- de Quervain’s thyroiditis, giant cell thyroiditis
- Fever, acute neck pain
- Tender goitre
- Leucocytosis with high ESR
- NSAIDS or corticosteroids in severe cases
- Histology showed the presence of giant cells and lymphocytes in the thyroid gland

# Congenital Hypothyroidism

- Local incidence 1:3000
- Neonatal cord blood TSH screening
- Suspicious if at 2 weeks:
  - fT4 <12 pmol/L and TSH >7 mIU/L
- Early treatment to preserve brain development
- Newborn
  - Cretinism, mental retardation, puffy face, deaf, mutism, protuberant abdomen, umbilical hernia
- Children
  - Retardation of growth, short stature

# Causes of Congenital Hypothyroidism

- Athyroesis
- Dyshormonogenesis
- Ectopic thyroid gland
- Associated with other congenital syndrome
  - Down syndrome, trisomy 18, congenital heart disease
- Hypothalamic pituitary hypothyroidism
- Transient hypothyroidism
  - Use of antithyroid drugs in mother
  - High anti-TSHR blocking Ab in mother
  - Neonatal iodine deficiency
  - Premature delivery

# Thyroxine Replacement

- With little residual thyroid function, replacement therapy requires approximately 1.6 microgram/kg of L-thyroxine daily
- For patients with known or suspected coronary heart disease
  - Hyperlipidaemia secondary to hypothyroidism predisposes to atherosclerosis
  - Initiation of T4 may raise cardiac output which further worsens cardiac symptoms and conditions
  - Manage coronary atherosclerosis prior to T4 supplementation
  - Usual starting dose 12.5 to 25 microgram daily
- TSH levels may decline within a month of initiating therapy
- Separate from interfering drugs at least 4 hours (e.g. iron / calcium-containing supplements)
- Changes in T4 doses need 4-6 weeks to equilibrate
- For women during pregnancy
  - Aim TSH $<2.5$ mIU/L (1st trimester), $<3.0$ (2nd trimester) and $<3.5$ mIU/L (3rd trimester)

# Myxoedematous Coma

- Medical emergency
- Severe hypothyroidism
- Confusion coma
- Hypothermia
- Respiratory failure
- Infection
- IV treatment of thyroxine
- IV hydrocortisone may be necessary

# Subclinical Hyperthyroidism

- Low TSH with normal fT4 levels
- Ix: Thyroid antibodies, ultrasound thyroid and/or thyroid scans
- Management:
  - Assess for development of thyrotoxic symptoms and complications (AF, osteoporosis)
  - Regular TFT monitoring

TABLE 10. SUBCLINICAL HYPERTHYROIDISM: WHEN TO TREAT

| *Factor* | *TSH (<0.1 mU/L)* | *TSH (0.1–0.4 mU/L)*[a] |
|---|---|---|
| Age >65 years | Yes | Consider treating |
| Age <65 years with comorbidities | | |
| Heart disease | Yes | Consider treating |
| Osteoporosis | Yes | Consider treating |
| Menopausal, not on estrogens or bisphosphonates | Yes | Consider treating |
| Hyperthyroid symptoms | Yes | Consider treating |
| Age <65 years, asymptomatic | Consider treating | Observe |

[a]Where 0.4 mU/L is the lower limit of the normal range.

ATA Guidelines Thyroid 2015

# Subclinical Hypothyroidism

- High TSH with normal fT4 levels
- Ix: Thyroid antibodies
- Management:
  - Assess lipid levels and hypothyroid symptoms
  - Regular TFT monitoring
  - Start T4 replacement if persistent TSH >10 mIU/L

# Iodine-Based Contrast Media-Induced Thyroid Dysfunction

- Can be hyperthyroidism or hypothyroidism
- Should be cautious in interpreting thyroid function if previous radiological examinations using iodine-based contrast media in the last 2-3 months
- Due to supraphysiological concentrations of iodine in the contrast solution
- Prevalence varies from 1-15%
- For hyperthyroidism, predominantly in regions with iodine deficiency and patients with underlying nodular goitre or latent Graves' disease
- For hypothyroidism, at-risk patients are those with autoimmune thyroiditis, living in areas with sufficient iodine supply
- Most cases mild and transient

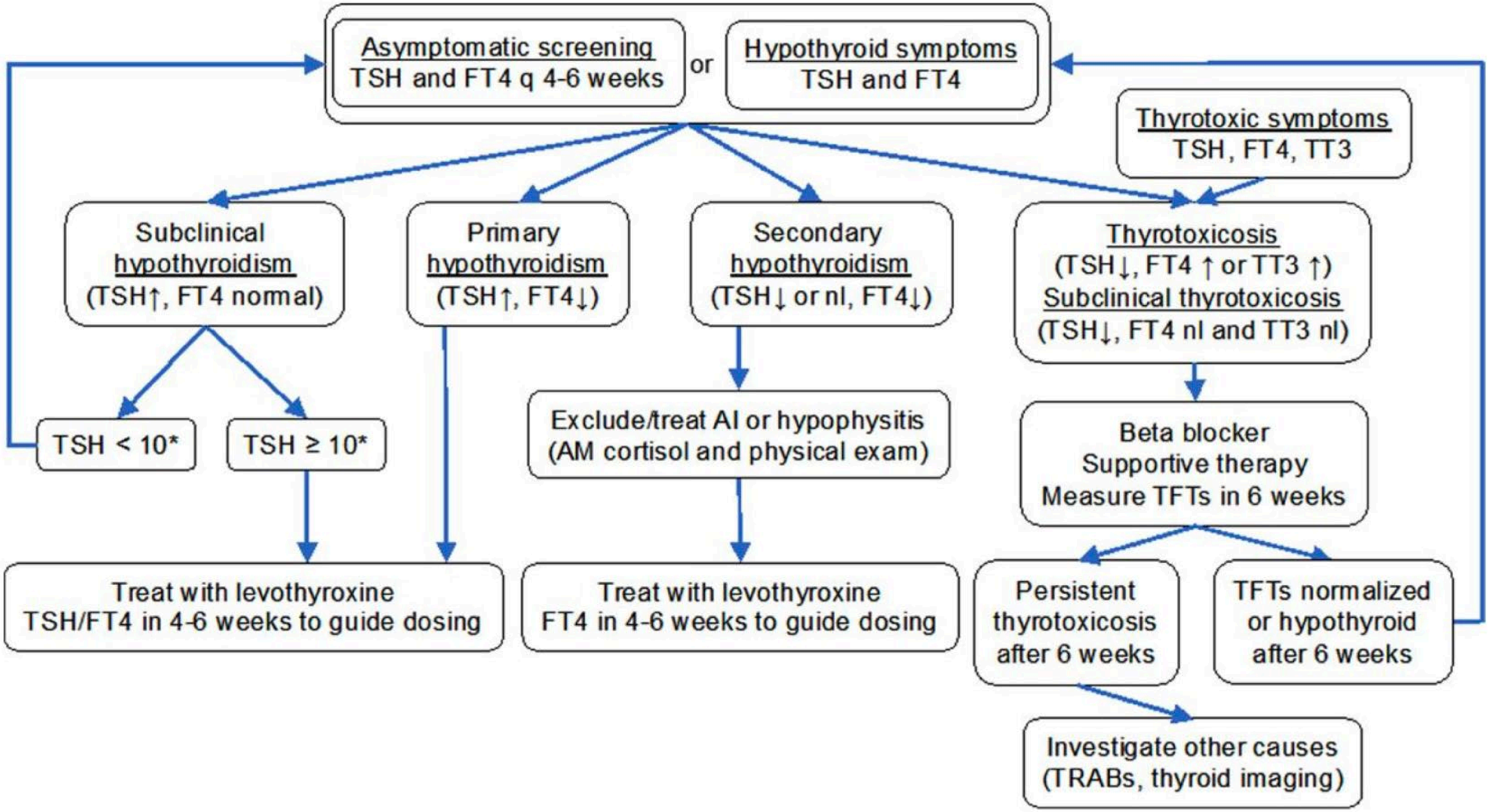
Eur Thyroid J 2021

# Immune Checkpoint Inhibitor-associated Thyroid Dysfunction

- The most common immunotherapy related adverse events (IRAE), can occur in as many as
- 25% of individuals treated with anti-CTLA4 monotherapy
- 38% of those treated with anti-PD-1/PD-L1 monotherapy
- 56% of those treated with dual anti-CTLA-4 and anti-PD-1/PD-L1 therapy
- Most cases develop within 6 weeks of initiation of immunotherapy, can also up to years into therapy
- Large majority of immunotherapy related thyrotoxicosis due to thyroiditis which is self-limiting
- Acute immune-mediated destruction of thyroid follicular cells
- Can evolve into hypothyroidism, or less commonly, resolve into euthyroid state within 6 weeks
- If central hypothyroidism is found, must exclude presence of concomitant secondary cortisol insufficiency

Wright JJ et al. J Clin Endocrinol Metab 2023

# Immune Checkpoint Inhibitor-associated Thyroid Dysfunction




Wright JJ et al. J Clin Endocrinol Metab 2023

MJA 2004; 180: 186–193

## 1: Interpretation matrix for thyroid function results*

| | High $T_4$ | Normal $T_4$ | Low $T_4$ |
|---|---|---|---|
| **High TSH** | In vivo or in vitro artefact<br>Pituitary hyperthyroidism [TSHoma]<br>Thyroid hormone resistance | Mild thyroid failure (primary)<br>(also termed subclinical hypothyroidism and diminished thyroid reserve) | Primary hypothyroidism |
| **Normal TSH** | As above<br>Sampling within 6 h of thyroxine dose | Normal (in patients taking thyroxine, TSH > 3 mU/L may indicate subtle underreplacement) | Pituitary or hypothalamic hypothyroidism<br>Severe non-thyroidal illness |
| **Low TSH** | Hyperthyroidism<br>(for this diagnosis, TSH must be suppressed rather than just low) | Subclinical hyperthyroidism<br>Subtle thyroxine overreplacement<br>Thyroid autonomy (multinodular goitre or autonomous functioning thyroid nodule)<br>Non-thyroidal illness | Pituitary or hypothalamic hypothyroidism<br>Severe non-thyroidal illness |

$T_4$ = serum free thyroxine. TSH = thyroid stimulating hormone.

* The Health Insurance Commission restricts financial reimbursement to serum TSH alone, except in specified clinical circumstances, when additional measurements of free $T_4$ and free triiodothyronine ($T_3$) are sanctioned. We suggest that practitioners routinely request "thyroid function tests" and provide clear clinical information to enable the laboratory to perform additional tests if justified. The information should include the suspected condition (especially if hypopituitarism) and medications (eg, thyroxine, carbimazole or propylthiouracil, amiodarone or phenytoin).

# Thank You!

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